



CSE Students



Placement-Focused



Hands-On

3-Day Hands-on Workshop on Agentic AI for Software Engineers

Build Intelligent AI Applications using
Python · LangChain · LangGraph · CrewAI · RAG

3

Intensive Days

9

Sessions

9+

Projects Built

100%

Hands-On

[View Full Curriculum ↓](#)



[Setup Checklist](#)



[Starter Requirements](#)

CONDUCTED BY



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Assistant Professor & AI Trainer · Dept. of CSE
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 What You'll Gain

Placement-Ready Outcomes

Leave with skills, projects, and confidence to crack AI-focused roles



Python + LLM Development

Strong foundation in Python-based AI application development with real LLM integrations



LangChain Expertise

Hands-on experience building chains, agents, and memory-powered applications



RAG Systems

Design and implement Retrieval-Augmented Generation pipelines from scratch



AI Agents

Build intelligent agents using ReAct pattern with web search, tools, and memory



Multi-Agent Systems

Create collaborative AI teams using CrewAI with specialized roles and workflows



GitHub Portfolio

Walk away with a capstone project and resume-worthy portfolio on GitHub



[Full Curriculum](#)

3-Day Workshop Schedule

A carefully designed progression from fundamentals to industry-level applications

Day 1

9:00 AM – 5:00 PM

Foundations of Generative AI & RAG Development

9:00 – 11:00

SESSION 1

Introduction to Generative AI & LLMs

📖 TOPICS COVERED

- › AI vs ML vs Deep Learning
- › Evolution of LLMs & Transformers
- › Prompt Engineering Fundamentals
- › APIs vs Local Models
- › What is Agentic AI?
- › AI Agents vs Agentic AI
- › Industry Use Cases & Career Opportunities

Python

Ollama

Llama 3

LLMs

⚡ HANDS-ON

- ✓ Install Python Environment & VS Code
- ✓ Ollama Installation & Setup
- ✓ Run Llama 3 Locally
- ✓ 🛠️ **Build your first AI Chatbot**

11:15 – 1:00

SESSION 2

LangChain Fundamentals

📖 TOPICS COVERED

- › LangChain Architecture Overview
- › Prompts & Prompt Templates
- › Chains & Chat Models
- › Output Parsers
- › Memory & Tools

⚡ HANDS-ON PROJECTS

📄 [AI Resume Reviewer](#)

✉️ [AI Email Generator](#)

Students learn prompt templates and structured outputs

2:00 – 5:00

SESSION 3

★ Flagship Project

Retrieval Augmented Generation (RAG)

📖 TOPICS COVERED

- › Why RAG? The Problem It Solves
- › Embeddings & Vector Databases
- › ChromaDB Setup & Usage
- › Document Loaders
- › Text Splitters & Chunking

⚡ CAPSTONE PROJECT

🏠 College Question Answering System

 Load PDFs → Create Vector DB → Ask Questions → Show Sources

PLACEMENT VALUE ★★★★★

RAG

ChromaDB

Embeddings

Vector DB

Day 2

9:00 AM – 5:00 PM

Autonomous & Multi-Agent Systems

9:00 – 11:00 AM

SESSION 1

Autonomous Single Agents with Tools (ReAct Loop)

📖 TOPICS COVERED

- › ReAct Architecture (Reason + Act Loop)
- › @tool Decorator & JSON Schema
- › Live Web Search & Wikimedia Integration
- › Safe Sandboxed Python Math Evaluator
- › Defensive Type-Handling for 3B LLMs

⚡ HANDS-ON PROJECTS



Project 1: Personal Research Assistant



Project 2: AI Fact-Checker & Web Detective

Live web search, Wikipedia queries, and safe math evaluations

LangGraph

ReAct

Tool Calling

Ollama

11:15 AM – 1:00 PM

SESSION 2

Self-Healing Agents & Code Execution

📖 TOPICS COVERED

- › Autonomous Code Generation & Parsing
- › Subprocess Isolation & Timeout Guards
- › Capturing Stderr Tracebacks Dynamically
- › Self-Correction & Reflection Loops
- › Iterative Prompt Feedback Chains

⚡ HANDS-ON PROJECT



Project 3: Self-Healing Python Coder



Write Code



Run Subprocess



Catch Stderr



Auto-Fix Loop

Agent iterates automatically until achieving exit code 0

Self-Healing

Subprocess

Debugging

LangChain

2:00 – 5:00 PM

SESSION 3

★ Multi-Agent Arena

Multi-Agent Collaboration, Debate & Reflection

📖 TOPICS COVERED

- › The Society of Mind Architecture
- › Adversarial Collaboration (Bull vs. Bear)
- › Evaluator-Optimizer Multi-Agent Pattern
- › Cross-Examination Message Passing
- › LLM-as-a-Judge Scoring & Feedback Loops

⚡ HANDS-ON PROJECTS

🔴 Project 4: AI Debate Arena

📝 Project 5: Writer & Strict Editor Duo

🟢 The Bull

🔴 The Bear

⚖️ Chief Judge

🖋️ Creator

🧐 Strict Editor

Multi-turn debate verdicts and quality score convergence ($\geq 8/10$)

AGENTIC DEPTH ★★★★★

Multi-Agent

Adversarial AI

Reflection

LLM Judge

Day 3

9:00 AM – 5:00 PM

Multi-Agent AI & Placement Project

9:00 – 11:00

SESSION 1

CrewAI — Multi-Agent Systems

📖 TOPICS COVERED

- › Multi-Agent System Architecture
- › Roles, Tasks & Crew Setup
- › Sequential Execution
- › Agent Collaboration Patterns

⚡ BUILD: AI SOFTWARE COMPANY



Project Manager



Researcher



Developer



Tester



Doc Writer

Students experience collaborative AI development

CrewAI

Multi-Agent

Collaboration

11:15 – 1:00

SESSION 2

Build an Industry-Level AI Application

🏗️ CHOOSE YOUR CAPSTONE PROJECT

01 AI Resume Analyzer

02 AI Interview Prep Assistant

03 College Admission Chatbot

04 Research Paper Assistant

05 AI Code Reviewer

06 AI Study Planner

👥 *Work in teams of 3–4 students*

Full Stack AI

Streamlit

Teamwork

2:00 – 5:00

 Grand Finale

Capstone Demo & Career Readiness

 EACH TEAM PRESENTS

- › Architecture Overview
- › AI Workflow Walkthrough
- › Live Demo
- › GitHub Repository

 CAREER GUIDANCE

- ✓ AI Interview Questions & Answers
- ✓ Resume Building for AI Roles
- ✓ GitHub Portfolio Best Practices
- ✓ 6-Month Learning Roadmap
- ✓ Open-Source Contributions
- ✓ Internship Opportunities
- ✓ AI Certifications Guide

Presentation

GitHub

Career

Portfolio

 COMPLEMENTARY MATERIAL

n8n — No-Code AI Workflow Automation



Build powerful AI automation pipelines visually — no backend code needed. Covers 4 complete projects: AI Email Responder, Document Summarizer, Support Webhook Bot, and Content Creation Pipeline.

 Email Auto-Responder Document Summarizer Support Bot Content Pipeline[Open n8n Material →](#)

 Tech Stack

Tools Used Throughout

Industry-standard tools that look great on your resume



Python



VS Code



Ollama



LangChain



LangGraph



CrewAI



ChromaDB



Streamlit



Git & GitHub

 Learning Journey

Your 3-Day Progression

A structured path from zero to industry-ready AI developer



DAY 1

Foundations & RAG

Set up environment, understand LLMs, build your first chatbot, and create a full RAG pipeline

 **Deliverable: College Q&A System**

DAY 2

Autonomous & Multi-Agent Systems

Single-agent ReAct tool calling, self-healing Python execution loops, adversarial debate arenas, and reflective creator-critic duos

 **Deliverable: 5 Complete Agentic AI Apps**

DAY 3

Multi-Agent Systems & Portfolio

Build an AI Software Company with CrewAI, develop capstone project, present and get career guidance

 **Deliverable: GitHub Portfolio Project**



Your Instructor

Conducted by



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 Jyothy Institute of Technology, Bengaluru

Passionate about making AI education practical and industry-relevant. Designing and delivering hands-on workshops that equip students with real-world skills in Generative AI, LangChain, RAG, and Multi-Agent Systems.

 Generative AI

 LangChain

 RAG Systems

 Multi-Agent

 Python

Agentic AI Workshop

Empowering engineering students to build the future with AI

#AgenticAI

#LangChain

#RAG

#CrewAI

#PlacementReady

Designed for maximum placement impact · 3 Days · 9 Sessions · 9+ Hands-On Projects